

IMAT QUESTION BANK: Maths, Physics and Logic

(Medium-Hard)

95 questions · 158 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · Question bank · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Algebra, functions and logarithms (Q1-16)

1. Given that $\log_{10} 2 \approx 0.30$ and $\log_{10} 3 \approx 0.48$, what is the value of $\log_{10} 7.5$?

- A) 0.18 B) 0.58 C) 0.88 D) 1.18 E) 1.88

Answer: C

$\log_{10} 7.5 = \log_{10}(15 \div 2) = \log_{10} 3 + \log_{10} 5 - \log_{10} 2$.

Since $\log_{10} 5 = 1 - \log_{10} 2 = 0.70$, the value is $0.48 + 0.70 - 0.30 = 0.88$

The answer 1.88 is $\log_{10} 75$, forgetting the division by 10, and 1.18 is $\log_{10} 15$, forgetting the division by 2.

2. Solve the equation $\log_3(x + 1) + \log_3(x - 1) = 1$.

- A) $x = \pm 2$ B) $x = 2$ only C) $x = -2$ only D) $x = 4$ E) $x = 1$

Answer: B

Combine the logarithms: $\log_3((x + 1)(x - 1)) = \log_3(x^2 - 1) = 1$, so $x^2 - 1 = 3$ and $x = \pm 2$

The arguments require $x > 1$, so the negative root is rejected and $x = 2$ is the only solution. The pair $x = \pm 2$ keeps the rejected root, for which $\log_3(x - 1)$ is not defined.

3. What is the value of $\log_8 32$?

- A) 2 B) $3/2$ C) 3 D) $4/3$ E) $5/3$

Answer: E

Write both arguments as powers of 2: $8 = 2^3$ and $32 = 2^5$

By change of base, $\log_8 32 = \log 32 \div \log 8 = 5 \log 2 \div (3 \log 2) = 5/3$. The value $3/2$ inverts the ratio, and 2 would solve $8^2 = 64$ rather than $8^x = 32$.

4. What is the solution of $\log_2 x + \log_4 x = 6$?

- A) 4 B) 8 C) 16 D) 32 E) 64

Answer: C

Change the base: $\log_4 x = (1/2) \log_2 x$, so the equation reads $(3/2) \log_2 x = 6$, hence $\log_2 x = 4$ and $x = 16$.

Check: $\log_2 16 + \log_4 16 = 4 + 2 = 6$

The answer 64 solves $\log_2 x = 6$ alone and drops the second term.

5. Which interval is the solution set of $(x - 1)(x + 2)(x - 4) > 0$?

- A) $-2 < x < 4$
- B) $1 < x < 4$
- C) $x < -2$ or $x > 4$
- D) $-2 < x < 1$ or $x > 4$
- E) $x < -2$ or $1 < x < 4$

Answer: D

The roots -2 , 1 and 4 split the number line into four intervals. Testing one point in each: for $x < -2$ the product is negative, for $-2 < x < 1$ it is positive, for $1 < x < 4$ it is negative, and for $x > 4$ it is positive. The solution is therefore $-2 < x < 1$ or $x > 4$. The pattern $x < -2$ or $x > 4$ tracks only the outer factors and ignores the middle root at $x = 1$.

6. What is the solution set of the inequality $(x - 3)/(x + 2) \leq 0$?

- A) $-2 < x \leq 3$
- B) $x \leq 3$
- C) $x < -2$ or $x \geq 3$
- D) $-2 \leq x \leq 3$
- E) $-2 < x < 3$

Answer: A

The fraction equals zero at $x = 3$, which is allowed by the $\leq$ sign, and is undefined at $x = -2$, which must be excluded. It is negative exactly between those values, where numerator and denominator have opposite signs. The solution is $-2 < x \leq 3$. The answer $x \leq 3$ ignores the sign change at $x = -2$, on both sides of which the fraction is positive.

7. What is the solution set of $|2x - 5| < 7$?

- A) $1 < x < 6$
- B) $-6 < x < -1$
- C) $-6 < x < 6$
- D) $-1 < x < 6$
- E) $x < 6$

Answer: D

$|2x - 5| < 7$ means $-7 < 2x - 5 < 7$. Adding 5 gives $-2 < 2x < 12$, so $-1 < x < 6$. The answer $1 < x < 6$ keeps only the branch $2x - 5 < 7$ and drops the branch $2x - 5 > -7$.

8. What is the solution set of the inequality $x^2 < x$?

- A) $0 < x < 1$
- B) $x < 0$
- C) $x < 0$ or $x > 1$
- D) $x < 1$
- E) $x > 1$

Answer: A

Rearranging: $x^2 - x < 0$, that is $x(x - 1) < 0$, which holds between the roots: $0 < x < 1$. The answer $x < 1$ comes from dividing by x without separating cases: for every $x < 0$ the left side is positive and the right side negative, so the inequality fails there.

9. Which expression is equal to $\sqrt{7 + 4\sqrt{3}}$?

- A) $4 + \sqrt{3}$
- B) $2 + \sqrt{3}$
- C) $\sqrt{7} + \sqrt{3}$
- D) $2\sqrt{3} + 1$
- E) $1 + \sqrt{6}$

Answer: B

Match the perfect square pattern $a^2 + 2ab + b^2$: $7 + 4\sqrt{3} = 2^2 + 2 \times 2 \times \sqrt{3} + (\sqrt{3})^2 = (2 + \sqrt{3})^2$, and since $2 + \sqrt{3} > 0$ the square root is $2 + \sqrt{3}$. As a check, $(1 + \sqrt{6})^2 = 7 + 2\sqrt{6}$, whose radical term does not match $4\sqrt{3}$.

10. What is the value of $27^{(2/3)} + 16^{(3/4)}$?

- A) 13 B) 17 C) 21 D) 26 E) 30

Answer: B

Take the root first, then the power: $27^{(2/3)} = (\sqrt[3]{27})^2 = 3^2 = 9$, and $16^{(3/4)} = (\sqrt[4]{16})^3 = 2^3 = 8$, so the sum is 17. The answer 30 multiplies base by exponent instead: $27 \times 2/3 + 16 \times 3/4 = 18 + 12$.

11. If $2^a = 3$ and $2^b = 5$, which of the following is equal to 60?

- A) $2^{(a+b)}$ B) $2^{(2a+b)}$ C) $2^{(a+b+2)}$ D) $2^{(a+2b)}$ E) $2^{(ab)}$

Answer: C

Factorise: $60 = 4 \times 3 \times 5$

With $2^a = 3$ and $2^b = 5$, this becomes $2^2 \times 2^a \times 2^b = 2^{(a+b+2)}$

The expression $2^{(ab)}$ multiplies the exponents, which is the rule for $(2^a)^b$, not for a product of powers.

12. For a school show, adult tickets cost 12 euro and child tickets cost 7 euro. In total 180 tickets were sold, and the takings were 1730 euro. How many adult tickets were sold?

- A) 84 B) 86 C) 90 D) 94 E) 96

Answer: D

Let a be adult and c child tickets: $a + c = 180$ and $12a + 7c = 1730$

Substituting $c = 180 - a$ gives $12a + 7(180 - a) = 1730$, so $5a = 470$ and $a = 94$

Check: $94 \times 12 + 86 \times 7 = 1128 + 602 = 1730$

The value 86 is the number of child tickets.

13. The sum of two numbers is 45 and their difference is 13. What is the larger number?

- A) 13 B) 16 C) 29 D) 32 E) 45

Answer: C

Adding the two conditions eliminates the smaller number: $45 + 13 = 58$ is twice the larger number, so the larger is 29 and the smaller 16. The value 16 is the smaller number itself, and $32 = 45 - 13$ is twice the smaller number.

14. A laboratory technician mixes a 20% saline solution with a 50% saline solution to obtain 6 dm³ of a 30% saline solution. What volume of the 50% solution is needed?

- A) 1 dm³ B) 2 dm³ C) 2.5 dm³ D) 3 dm³ E) 4 dm³

Answer: B

Let y be the volume of the 50% solution.

Salt balance: $0.20(6 - y) + 0.50y = 0.30 \times 6 = 1.8$, so $1.2 + 0.3y = 1.8$ and $y = 2$ dm³

The value 4 dm³ is the volume of the 20% solution used.

15. The function f is defined by $f(x) = (x + 1)/(x - 1)$ for $x \neq 1$. What is $f(f(3))$?

- A) 1/2 B) 2 C) 3 D) 4 E) 9

Answer: C

Step 1: $f(3) = (3 + 1)/(3 - 1) = 4/2 = 2$

Step 2: $f(2) = (2 + 1)/(2 - 1) = 3$

So $f(f(3)) = 3$ (this function is its own inverse). The value 2 stops after a single application of f .

16. The function f is defined by $f(x) = (2x + 1)/(x - 3)$ for $x \neq 3$. Which of the following is the inverse function $f^{-1}(x)$?

- A)** $(3x - 1)/(x + 2)$
- B)** $(x + 3)/(2x - 1)$
- C)** $(2x + 1)/(x - 3)$
- D)** $(3x + 1)/(x + 2)$
- E)** $(3x + 1)/(x - 2)$

Answer: E

Set $y = (2x + 1)/(x - 3)$ and solve for x : $y(x - 3) = 2x + 1$, so $xy - 3y = 2x + 1$, hence $x(y - 2) = 3y + 1$ and $x = (3y + 1)/(y - 2)$

Therefore $f^{-1}(x) = (3x + 1)/(x - 2)$

The expression $(2x + 1)/(x - 3)$ is f itself, left unchanged instead of inverted.

Section B - Geometry, trigonometry and graphs (Q17-28)

17. A solid wooden block has the shape of a right circular cylinder of radius 6 cm and height 9 cm. A conical hole with the same base radius and the same height is drilled out of the block. What fraction of the cylinder's volume remains?

- A)** $1/3$ **B)** $1/2$ **C)** $2/3$ **D)** $3/4$ **E)** $8/9$

Answer: C

The cone shares the cylinder's base and height, so its volume is one third of the cylinder's: $(1/3)\pi r^2 h$ against $\pi r^2 h$.

What remains is $1 - 1/3 = 2/3$ of the volume

The fraction $1/3$ is the part removed, not the part left.

18. A circle of radius 5 cm is inscribed in a square. Taking $\pi = 3.14$, what is the area of the part of the square that lies outside the circle?

- A)** 21.5 cm^2 **B)** 39.25 cm^2 **C)** 78.5 cm^2 **D)** 100 cm^2 **E)** 121.5 cm^2

Answer: A

An inscribed circle touches all four sides, so the square has side $2 \times 5 = 10 \text{ cm}$ and area 100 cm^2 .

The circle's area is $3.14 \times 5^2 = 78.5 \text{ cm}^2$

Outside area = $100 - 78.5 = 21.5 \text{ cm}^2$

The value 121.5 cm^2 adds the two areas instead of subtracting them.

19. An equilateral triangle has sides of length 12 cm. What is its area?

- A)** $36\sqrt{3} \text{ cm}^2$ **B)** 72 cm^2 **C)** $72\sqrt{3} \text{ cm}^2$ **D)** 144 cm^2 **E)** $144\sqrt{3} \text{ cm}^2$

Answer: A

The height comes from the 30-60-90 half triangle: $h = 12 \times (\sqrt{3}/2) = 6\sqrt{3} \text{ cm}$.

Area = $(1/2) \times 12 \times 6\sqrt{3} = 36\sqrt{3} \text{ cm}^2$

The value 72 cm^2 uses the side length 12 as if it were the height.

20. A closed cylindrical tin has radius 3 cm and height 8 cm. What is its total surface area, ends included?

- A)** $42\pi \text{ cm}^2$ **B)** $48\pi \text{ cm}^2$ **C)** $57\pi \text{ cm}^2$ **D)** $66\pi \text{ cm}^2$ **E)** $72\pi \text{ cm}^2$

Answer: D

Total surface area = two ends plus curved wall = $2\pi r^2 + 2\pi rh = 2\pi \times 9 + 2\pi \times 24 = 18\pi + 48\pi = 66\pi \text{ cm}^2$. The value $48\pi \text{ cm}^2$ counts only the wall, and $57\pi \text{ cm}^2$ forgets one of the two ends.

21. What is the angle between the hour hand and the minute hand of a clock at 3:40? (The hour hand turns at 0.5° per minute, the minute hand at 6° per minute.)

- A)** 100° **B)** 110° **C)** 130° **D)** 150° **E)** 230°

Answer: C

Measured from 12: hour hand at $3 \times 30^\circ + 40 \times 0.5^\circ = 110^\circ$, minute hand at $40 \times 6^\circ = 240^\circ$.

Angle between them: $240^\circ - 110^\circ = 130^\circ$

The answer 150° freezes the hour hand at exactly 3 o'clock and ignores its 20° drift during the 40 minutes.

22. Between 7:00 and 8:00 the hands of a clock are exactly 100 degrees apart at 7:20. At what later time in that hour are they exactly 100 degrees apart again? (The hour hand turns at 0.5 degrees per minute, the minute hand at 6 degrees per minute.)

- A)** 7:52 and 8/11 minutes
B) 7:54 and 6/11 minutes
C) 7:56 and 4/11 minutes
D) 7:58 and 2/11 minutes
E) 8:00 exactly

Answer: C

Let m be minutes past 7:00. Hour hand at $210 + 0.5m$, minute hand at $6m$, so their separation is $|5.5m - 210|$. Setting $5.5m - 210 = 100$ gives $m = 310 \div 5.5 = 56$ and $4/11$ minutes, that is 7:56 and $4/11$ (the earlier root $m = 20$ is the given 7:20). The time 7:52 and $8/11$ answers a separation of 80° instead of 100° , and 8:00 exactly assumes the hands approach at 5° per minute instead of 5.5° .

23. What is the value of $\tan 30^\circ \times \tan 60^\circ$?

- A)** 3 **B)** $\sqrt{3}/3$ **C)** $\sqrt{3}$ **D)** $1/3$ **E)** 1

Answer: E

$\tan 30^\circ = \sqrt{3}/3$ and $\tan 60^\circ = \sqrt{3}$, so the product is $(\sqrt{3}/3) \times \sqrt{3} = 3/3 = 1$ (complementary angles have tangents that are reciprocals). The value $1/3$ uses $\tan 30^\circ$ twice, and 3 divides the two values instead of multiplying them.

24. For an angle θ , what is the value of $(\sin \theta + \cos \theta)^2 - 2 \sin \theta \cos \theta$?

- A)** $\sin^2 \theta - \cos^2 \theta$ **B)** 2 **C)** 0 **D)** $2 \sin \theta \cos \theta$ **E)** 1

Answer: E

Expand: $(\sin \theta + \cos \theta)^2 = \sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta$. Subtracting $2 \sin \theta \cos \theta$ cancels the middle term exactly, leaving $\sin^2 \theta + \cos^2 \theta = 1$. The expression $2 \sin \theta \cos \theta$ stops after the expansion and forgets the cancellation.

25. What is the value of $(\sin 45^\circ - \cos 45^\circ)^2 + 2 \sin 30^\circ \cos 60^\circ$?

- A) 0 B) $1/4$ C) $1/2$ D) 1 E) $3/4$

Answer: C

$\sin 45^\circ = \cos 45^\circ$, so the squared term is 0

Then $2 \sin 30^\circ \cos 60^\circ = 2 \times (1/2) \times (1/2) = 1/2$, and the total is $1/2$

The answer 0 forgets the second term altogether.

26. A straight line passes through the points (2, 5) and (6, 17). What is its slope?

- A) $1/4$ B) $1/3$ C) 3 D) 4 E) 12

Answer: C

Slope = change in y over change in x = $(17 - 5) \div (6 - 2) = 12 \div 4 = 3$

The values 4 and 12 each use only one of the two changes, and $1/3$ inverts the ratio.

27. A cyclist rides along a straight road at a constant speed. Her distance from the start is recorded as 1.8 km after 6 minutes and 3.0 km after 10 minutes. What is her speed in m/s?

- A) 3 m/s B) 5 m/s C) 18 m/s D) 50 m/s E) 300 m/s

Answer: B

Between the two records she covers $3.0 - 1.8 = 1.2$ km = 1200 m in $10 - 6 = 4$ min = 240 s.

Speed = $1200 \div 240 = 5$ m/s

The value 18 m/s is her speed in km/h (0.3 km per minute $\times 60$) left unconverted, and 300 m/s keeps the rate in metres per minute.

28. Water is pumped into a tank that already contains 40 L. The volume reaches 90 L after 5 minutes and 160 L after 15 minutes. What is the average rate of inflow over the whole 15 minutes?

- A) 7 L/min B) 8 L/min C) 8.5 L/min D) 10 L/min E) 160 L/min

Answer: B

Average rate = total change $\div$ total time = $(160 - 40) \div 15 = 120 \div 15 = 8$ L/min. The value 8.5 L/min averages the two segment rates (10 and 7 L/min) instead of dividing the total change by the total time.

Section C - Probability and combinatorics (Q29-36)

29. Two fair six-sided dice are rolled once. What is the probability that the sum of the two scores is at least 9?

- A) $1/9$ B) $1/3$ C) $5/36$ D) $5/18$ E) $5/12$

Answer: D

Counting ordered outcomes: sum 9 in 4 ways, sum 10 in 3, sum 11 in 2, sum 12 in 1, so 10 of 36.

$P = 10/36 = 5/18$

The value $5/36$ counts only the five unordered pairs and forgets that (4,6) and (6,4) are different rolls; the value $1/9$ answers a sum of exactly 9.

30. Three fair six-sided dice are rolled. What is the probability that all three show the same number?

- A)** $1/216$ **B)** $1/108$ **C)** $1/72$ **D)** $1/36$ **E)** $1/6$

Answer: D

Let the first die show anything; each of the other two must then match it: $P = 1 \times (1/6) \times (1/6) = 1/36$. The value $1/216$ fixes all three faces in advance, counting only one specific triple instead of all six matching triples, and $1/6$ is the chance of three copies of one particular named number.

31. Two fair six-sided dice are rolled. What is the probability that the two scores are a 2 and a 5, in either order?

- A)** $1/36$ **B)** $1/18$ **C)** $1/9$ **D)** $1/6$ **E)** $1/3$

Answer: B

The favourable ordered outcomes are (2,5) and (5,2), so $P = 2/36 = 1/18$

The value $1/36$ takes a single ordering, and $1/9$ wrongly counts four outcomes by also including the doubles (2,2) and (5,5).

32. Two fair six-sided dice are rolled. What is the probability that the sum is 7 or at least one die shows a six?

- A)** $1/6$ **B)** $5/12$ **C)** $7/18$ **D)** $11/36$ **E)** $17/36$

Answer: B

$P(\text{sum } 7) = 6/36$ and $P(\text{at least one six}) = 11/36$, but the rolls (1,6) and (6,1) satisfy both and are counted twice.

Inclusion-exclusion gives $(6 + 11 - 2)/36 = 15/36 = 5/12$

The value $17/36$ adds the two probabilities without subtracting the overlap.

33. A fair coin is flipped four times. What is the probability of getting exactly two heads?

- A)** $1/8$ **B)** $1/4$ **C)** $1/2$ **D)** $3/16$ **E)** $3/8$

Answer: E

There are $2^4 = 16$ equally likely head-tail sequences, and $C(4,2) = 6$ of them contain exactly two heads, so $P = 6/16 = 3/8$. The value $1/4$ counts only some of the six arrangements.

34. An urn contains 3 red, 4 green and 5 blue balls. Two balls are drawn at random without replacement. What is the probability that at least one of them is green?

- A)** $1/3$ **B)** $3/4$ **C)** $14/33$ **D)** $16/33$ **E)** $19/33$

Answer: E

Use the complement: $P(\text{no green}) = (8/12) \times (7/11) = 56/132 = 14/33$, so $P(\text{at least one green}) = 1 - 14/33 = 19/33$. The value $16/33$ answers exactly one green, and $14/33$ is the complement itself.

35. The probability that a candidate passes a maths test is $2/3$, and the probability that she passes an independent logic test is $3/4$. What is the probability that she passes exactly one of the two tests?

- A)** $1/2$ **B)** $1/6$ **C)** $1/12$ **D)** $5/12$ **E)** $11/12$

Answer: D

Exactly one pass = (pass maths and fail logic) or (fail maths and pass logic): $(2/3 \times 1/4) + (1/3 \times 3/4) = 2/12 + 3/12 = 5/12$. The value $1/2$ is the probability of passing both, and $11/12$ is the probability of passing at least one.

36. A fair coin is tossed and a fair six-sided die is rolled. What is the probability of obtaining tails and a score greater than 4?

- A)** 1/12 **B)** 1/6 **C)** 1/4 **D)** 1/2 **E)** 2/3

Answer: B

The events are independent: $P(\text{tails}) \times P(\text{score above 4}) = (1/2) \times (2/6) = 1/6$. The value 2/3 computes the union of the two events instead of their joint occurrence, and 1/12 uses a single high face instead of the two faces 5 and 6.

Section D - Unit conversion and arithmetic (Q37-40)

37. A high-speed train covers 270 km in 3.0 h at a constant speed. Expressed in m/s, its speed is:

- A)** 25 m/s **B)** 75 m/s **C)** 90 m/s **D)** 324 m/s **E)** 1500 m/s

Answer: A

Speed = $270 \div 3.0 = 90$ km/h

Converting: $90 \text{ km/h} = 90\,000 \text{ m in } 3600 \text{ s} = 25 \text{ m/s}$ (divide km/h by 3.6)

The value 324 m/s multiplies by 3.6 instead of dividing, and 1500 m/s gives metres per minute rather than metres per second.

38. A swimming pool is 5.0 m long, 2.0 m wide and 1.2 m deep. What volume of water does it hold, expressed in litres?

- A)** 12 L **B)** 120 L **C)** 1200 L **D)** 12000 L **E)** 120000 L

Answer: D

$V = 5.0 \times 2.0 \times 1.2 = 12 \text{ m}^3$

Since $1 \text{ m}^3 = 1000 \text{ dm}^3$ and $1 \text{ dm}^3 = 1 \text{ L}$, the pool holds $12 \times 1000 = 12\,000 \text{ L}$

The value 1200 L multiplies by 100 instead of 1000, confusing a volume factor with a length factor.

39. A water tank is $2/3$ full. A valve then drains $1/4$ of the tank's total capacity. What fraction of the tank is still full?

- A)** 1/6 **B)** 1/4 **C)** 1/2 **D)** 5/12 **E)** 11/12

Answer: D

Draining removes $1/4$ of the capacity, so the remaining fraction is $2/3 - 1/4 = 8/12 - 3/12 = 5/12$. The value $1/2$ instead removes a quarter of the water already present ($2/3 \times 3/4$), and $11/12$ adds the fractions instead of subtracting.

40. A water tank is one third full. After 40 more litres are poured in, the tank is three quarters full. What is the full capacity of the tank?

- A)** 60 litres **B)** 72 litres **C)** 90 litres **D)** 96 litres **E)** 120 litres

Answer: D

The 40 L fill the fraction $3/4 - 1/3 = 9/12 - 4/12 = 5/12$ of the capacity, so the capacity is $40 \times 12/5 = 96 \text{ L}$

The value 120 L treats 40 L as one third of the tank, and 60 L treats the 40 L as the two thirds needed to go from one third full to completely full.

Section E - Kinematics and dynamics (Q41-56)

41. The velocity of a trolley moving along a straight track is recorded every 2 s: at 0 s it is 0 m/s, at 2 s it is 6 m/s, at 4 s it is 10 m/s, at 6 s it is 12 m/s and at 8 s it is 12 m/s. Which statement is correct?

- A)** Its acceleration between 4 s and 6 s is 1 m/s^2 .
- B)** Its average velocity over the whole 8 s is 10 m/s.
- C)** It decelerates between 6 s and 8 s.
- D)** Its acceleration is constant at 3 m/s^2 between 0 s and 6 s.
- E)** It travels 96 m during the 8 s.

Answer: A

Successive accelerations: $(6 - 0)/2 = 3$, $(10 - 6)/2 = 2$, $(12 - 10)/2 = 1$ and $(12 - 12)/2 = 0 \text{ m/s}^2$, so the acceleration is not constant at 3 m/s^2 over the first 6 s. Displacement is the area under the velocity-time graph: $6 + 16 + 22 + 24 = 68 \text{ m}$, so the average velocity is $68 \div 8 = 8.5 \text{ m/s}$ and the distance is not 96 m. From 6 s to 8 s the velocity is constant at 12 m/s, so there is no deceleration. The correct statement is that the acceleration between 4 s and 6 s is $(12 - 10) \div 2 = 1 \text{ m/s}^2$.

42. In a race, a sprinter increases her speed steadily from 0 to 8.0 m/s during the first 4.0 s and then keeps running at a constant 8.0 m/s for the next 6.0 s. What is her average speed over the whole 10 s?

- A)** 1.6 m/s **B)** 4.0 m/s **C)** 4.8 m/s **D)** 6.4 m/s **E)** 8.0 m/s

Answer: D

Distance = area under the velocity-time graph: $(1/2) \times 4.0 \times 8.0 = 16 \text{ m}$ while accelerating, plus $8.0 \times 6.0 = 48 \text{ m}$ at constant speed, total 64 m.

Average speed = $64 \div 10 = 6.4 \text{ m/s}$

The value 8.0 m/s takes the final speed as the whole-run average, and 4.0 m/s would hold only if the acceleration lasted all 10 s.

43. On a straight air track, a glider passes sensor P moving at 2.0 m/s and sensor Q moving at 4.0 m/s. The sensors are 6.0 m apart and the glider takes 2.0 s to travel from P to Q. Which statement is correct?

- A)** Its average velocity between the gates is 3.0 m/s and its acceleration is 2.0 m/s^2 .
- B)** Its average velocity between the gates is 2.0 m/s and its acceleration is 1.0 m/s^2 .
- C)** Its average velocity between the gates is 4.0 m/s and its acceleration is 1.0 m/s^2 .
- D)** Its average velocity between the gates is 6.0 m/s and its acceleration is 1.0 m/s^2 .
- E)** Its average velocity between the gates is 3.0 m/s and its acceleration is 1.0 m/s^2 .

Answer: E

Average velocity = displacement $\div$ time = $6.0 \div 2.0 = 3.0 \text{ m/s}$

Acceleration = change in velocity $\div$ time = $(4.0 - 2.0) \div 2.0 = 1.0 \text{ m/s}^2$

The value 2.0 m/s^2 divides the speed change by 1 s instead of 2 s, and averages of 2.0 or 4.0 m/s borrow one instantaneous gate speed for the whole run.

44. Which of the following statements about motion graphs is correct?

- A)** For uniformly accelerated motion starting from rest, the average velocity during the acceleration is half the final velocity.
- B)** A curved velocity-time graph means the object is moving along a circular path.
- C)** The average velocity over any time interval always equals the mean of the initial and final velocities.
- D)** The area under a velocity-time graph gives the acceleration.
- E)** The gradient of a velocity-time graph gives the distance travelled.

Answer: A

The gradient of a velocity-time graph gives acceleration, not distance, and the area gives displacement, not acceleration. The mean of the initial and final velocities equals the average velocity only when the acceleration is uniform. For uniform acceleration from rest, $v = at$, so the average velocity $(1/2)at$ is exactly half the final velocity at , which makes that the correct statement.

45. Which statement correctly distinguishes the tangential and the centripetal components of acceleration for an object moving in a circle?

- A)** The tangential component acts towards the centre, and the centripetal component acts along the tangent.
- B)** Both components change only the direction of motion.
- C)** Both components change only the speed of the object.
- D)** The centripetal component changes the speed, and the tangential component changes the direction of motion.
- E)** The tangential component changes the speed, and the centripetal component changes the direction of the velocity.

Answer: E

Tangential acceleration acts along the tangent and changes the speed, the magnitude of the velocity. Centripetal acceleration acts towards the centre and changes only the direction of the velocity. The version that swaps the two roles reverses their definitions, and any claim that both components change only the speed, or only the direction, contradicts what each component does.

46. A cyclist rides around a circular track of radius 20 m at a constant speed of 10 m/s. Which statement about her acceleration is correct?

- A)** It is 5.0 m/s², directed towards the centre of the track.
- B)** It is 2.0 m/s², directed away from the centre of the track.
- C)** It is 5.0 m/s², directed along her direction of travel.
- D)** It is 0.50 m/s², directed along her direction of travel.
- E)** It is zero, because her speed is constant.

Answer: A

Constant speed on a circle still means a continuously changing direction, so the acceleration is purely centripetal: $a = v^2/r = 10^2 \div 20 = 5.0 \text{ m/s}^2$, directed towards the centre. Zero acceleration ignores the direction change, and the values 0.50 m/s² and 2.0 m/s² come from v/r and r/v instead of v^2/r .

47. A car is driven around a circular test track while the driver steadily presses the accelerator, so the car is speeding up. At such an instant, which statement about the car's acceleration is correct?

- A)** It has a component along the motion and a component towards the centre, so the resultant points forwards and inwards.
- B)** It points directly towards the centre of the track.
- C)** It is zero, because the tangential and centripetal parts cancel.
- D)** It points directly away from the centre, in the direction of the outward push the driver feels.
- E)** It points along the direction of motion, since only a change of speed can produce acceleration.

Answer: A

Because the speed changes there is a tangential component along the direction of motion; because the direction changes there is a centripetal component towards the centre. The resultant therefore points forwards and inwards, between tangent and radius. Pointing straight at the centre would require constant speed, and a resultant pointing outwards never occurs in circular motion.

48. A car travels at 72 km/h along a straight road when the driver sees a hazard. Her reaction time is 1.0 s, after which the brakes produce a constant deceleration of 5.0 m/s² until the car stops. What total distance does the car cover between the hazard being seen and the car stopping?

- A)** 20 m **B)** 40 m **C)** 60 m **D)** 80 m **E)** 100 m

Answer: C

Convert: 72 km/h = $72 \div 3.6 = 20$ m/s

Reaction distance: $20 \times 1.0 = 20$ m

Braking distance from $v^2 = 2as$: $s = 20^2 \div (2 \times 5.0) = 40$ m, equivalently average braking speed 10 m/s over 4.0 s.

Total = $20 + 40 = 60$ m

The value 40 m forgets the reaction phase, and 80 m multiplies the full 20 m/s by the whole braking time instead of using the average 10 m/s.

49. A train travelling at 30 m/s applies its brakes and comes to rest after covering 150 m at constant deceleration. What is the magnitude of the deceleration?

- A)** 0.20 m/s² **B)** 1.5 m/s² **C)** 3.0 m/s² **D)** 5.0 m/s² **E)** 6.0 m/s²

Answer: C

With final speed zero in $v^2 = u^2 + 2as$: $a = 30^2 \div (2 \times 150) = 900 \div 300 = 3.0$ m/s²

The value 6.0 m/s² drops the factor of 2, and 1.5 m/s² squares the average speed of 15 m/s instead of the initial speed.

50. A crate of mass 4.0 kg is pulled along a horizontal floor by a horizontal force of 24 N. A frictional force of 8.0 N opposes the motion. What is the acceleration of the crate?

- A)** 2.0 m/s² **B)** 4.0 m/s² **C)** 6.0 m/s² **D)** 8.0 m/s² **E)** 16 m/s²

Answer: B

Resultant force = $24 - 8.0 = 16$ N, since friction opposes the pull.

Newton's second law: $a = 16 \div 4.0 = 4.0$ m/s²

The value 6.0 m/s² divides the pull by the mass and ignores friction, and 8.0 m/s² quotes the resultant force in newtons as if it were an acceleration.

51. A block of mass 2.0 kg slides down a frictionless slope inclined at 30° to the horizontal. What is its acceleration along the slope? ($g = 10 \text{ m/s}^2$, $\sin 30^\circ = 0.50$, $\cos 30^\circ = 0.87$.)

- A)** 5.0 m/s² **B)** 8.7 m/s² **C)** 10 m/s² **D)** 17.4 m/s² **E)** 20 m/s²

Answer: A

Only the weight component along the slope accelerates the block: $F = mg \sin 30^\circ = 2.0 \times 10 \times 0.50 = 10 \text{ N}$, so $a = 10 \div 2.0 = 5.0 \text{ m/s}^2$ (equivalently $a = g \sin 30^\circ$; the mass cancels). The value 8.7 m/s² uses $\cos 30^\circ$, the component perpendicular to the slope, and 17.4 or 20 m/s² quote forces in newtons as accelerations without dividing by the mass.

52. A crane lifts a 50 kg load by applying a vertical upward force of 550 N. What is the upward acceleration of the load? ($g = 10 \text{ m/s}^2$.)

- A)** 0.10 m/s² **B)** 1.0 m/s² **C)** 10 m/s² **D)** 11 m/s² **E)** 21 m/s²

Answer: B

Weight = $50 \times 10 = 500 \text{ N}$

Resultant upward force = $550 - 500 = 50 \text{ N}$, so $a = 50 \div 50 = 1.0 \text{ m/s}^2$

The value 11 m/s² ignores the weight entirely, and 21 m/s² adds the two forces although they act in opposite directions.

53. An astronaut of mass 60 kg floats at rest in space and throws a tool of mass 2.0 kg so that it moves away from her at 6.0 m/s. What is the velocity of the astronaut immediately afterwards?

- A)** 0.20 m/s in the direction of the throw.
B) 0.20 m/s opposite to the direction of the throw.
C) 0.40 m/s opposite to the direction of the throw.
D) 6.0 m/s opposite to the direction of the throw.
E) 12 m/s opposite to the direction of the throw.

Answer: B

Momentum is conserved from zero total: the tool carries $2.0 \times 6.0 = 12 \text{ kg}\cdot\text{m/s}$ in the direction of the throw, so the astronaut gains 12 kg·m/s the opposite way.

Her speed is $12 \div 60 = 0.20 \text{ m/s}$, opposite to the throw

The value 12 m/s reads the momentum value as a velocity, and 6.0 m/s assumes her speed equals the tool's.

54. A ball of mass 0.50 kg strikes a wall horizontally at 12 m/s and rebounds along its original line at 8.0 m/s. The contact with the wall lasts 0.020 s. What is the magnitude of the average force exerted by the wall on the ball?

- A)** 100 N **B)** 200 N **C)** 300 N **D)** 500 N **E)** 1000 N

Answer: D

The ball reverses direction, so the momentum change has magnitude $0.50 \times (12 + 8.0) = 10 \text{ kg}\cdot\text{m/s}$.

Average force = $\Delta p \div \Delta t = 10 \div 0.020 = 500 \text{ N}$

The value 100 N subtracts the speeds as if the ball continued through the wall without rebounding, and 1000 N adds the speeds but forgets the mass.

55. A motor lifts a 60 kg load at constant speed through a height of 10 m in 20 s. What is the useful output power of the motor? ($g = 10 \text{ m/s}^2$.)

- A)** 30 W **B)** 150 W **C)** 300 W **D)** 600 W **E)** 6000 W

Answer: C

At constant speed the lifting force equals the weight, $60 \times 10 = 600 \text{ N}$.

$$\text{Work} = \text{force} \times \text{distance} = 600 \times 10 = 6000 \text{ J}$$

$$\text{Power} = 6000 \div 20 = 300 \text{ W}$$

The value 30 W leaves g out of the work, and 6000 W quotes the work in joules as if it were a power.

56. A car moves at a constant 20 m/s along a horizontal road. The total resistive force on it is 600 N. What power must the engine deliver to maintain this speed?

- A)** 30 W **B)** 600 W **C)** 6000 W **D)** 12000 W **E)** 24000 W

Answer: D

At constant speed the driving force balances the 600 N resistance.

$$\text{Power} = Fv = 600 \times 20 = 12\,000 \text{ W}$$

The value 30 W divides force by speed instead of multiplying, and 600 W is just the force itself rather than force times velocity.

Section F - Fluids, thermal physics and electricity (Q57-75)

57. A diver works at a depth of 15 m in a fresh-water lake. The density of water is 1000 kg/m^3 , $g = 10 \text{ m/s}^2$, and the atmospheric pressure at the surface is 100 kPa. Which statement about the pressures on the diver is correct?

- A)** The absolute pressure at 15 m is 150 kPa, because atmospheric pressure does not act underwater.
B) The gauge pressure at 15 m is 250 kPa, and the absolute pressure is 350 kPa.
C) The gauge pressure at 15 m is 150 kPa, and the absolute pressure is 50 kPa.
D) The gauge pressure at 15 m is 1.5 kPa, and the absolute pressure is 101.5 kPa.
E) The gauge pressure at 15 m is 150 kPa, and the absolute pressure is 250 kPa.

Answer: E

Stevin's law gives the gauge pressure: $\rho gh = 1000 \times 10 \times 15 = 150\,000 \text{ Pa} = 150 \text{ kPa}$. The absolute pressure adds the atmospheric pressure acting on the surface: $150 + 100 = 250 \text{ kPa}$. The claim that the atmosphere does not act underwater is wrong: the water column rests on the surface, so the atmospheric push is transmitted down through the water.

58. An open-tube manometer contains mercury of density 13600 kg/m^3 . The mercury level in the arm open to the atmosphere stands 100 mm above the level in the arm connected to a gas supply. Atmospheric pressure is 100 kPa and $g = 10 \text{ m/s}^2$. What is the absolute pressure of the gas supply?

- A)** 13.6 kPa **B)** 86.4 kPa **C)** 100 kPa **D)** 101 kPa **E)** 113.6 kPa

Answer: E

The level in the open arm stands higher, so the gas pressure exceeds atmospheric by $\rho gh = 13600 \times 10 \times 0.100 = 13\,600 \text{ Pa} = 13.6 \text{ kPa}$.

$$\text{Absolute pressure} = 100 + 13.6 = 113.6 \text{ kPa}$$

The value 86.4 kPa subtracts the column instead of adding it, and 101 kPa computes the column with the density of water (about 1 kPa) instead of mercury.

59. A tank contains 0.80 m of oil of density 800 kg/m³ floating on 1.2 m of water of density 1000 kg/m³. Taking $g = 10 \text{ m/s}^2$ what is the gauge pressure at the bottom of the tank?

- A)** 1.8 kPa **B)** 6.4 kPa **C)** 12 kPa **D)** 18.4 kPa **E)** 20 kPa

Answer: D

Each layer adds its own ρgh : oil gives $800 \times 10 \times 0.80 = 6.4 \text{ kPa}$ and water gives $1000 \times 10 \times 1.2 = 12 \text{ kPa}$.

Total gauge pressure = 18.4 kPa

The value 20 kPa uses the density of water for both layers, and 12 kPa forgets the oil layer.

60. Three blocks are placed in oil of density 800 kg/m³. Block P floats with exactly half its volume submerged, block Q floats with one quarter of its volume submerged, and block R sinks and rests on the bottom. Which statement is correct?

- A)** Q has a density of about 200 kg/m³, the lowest of the three, and R is denser than the oil.
B) R has a density of exactly 800 kg/m³, the same as the oil.
C) P and Q must have the same density, because both are floating.
D) Q is denser than P, because a smaller fraction of Q is submerged.
E) P is the densest of the three blocks.

Answer: A

A floating block displaces its own weight, so its density is the submerged fraction times the oil's density: block P: $0.5 \times 800 = 400 \text{ kg/m}^3$; block Q: $0.25 \times 800 = 200 \text{ kg/m}^3$. A block resting on the bottom must be denser than the fluid, so R exceeds 800 kg/m³. Hence Q at 200 kg/m³ is the least dense and R is denser than the oil. The claim that Q is denser than P inverts the rule: the smaller the submerged fraction, the lower the density.

61. A block of weight 6.0 N and density 750 kg/m³ floats freely in water of density 1000 kg/m³. What is the apparent weight of the block while it floats?

- A)** 0 N **B)** 1.5 N **C)** 2.0 N **D)** 4.5 N **E)** 6.0 N

Answer: A

A floating block is in equilibrium: upthrust = weight = 6.0 N exactly (it displaces its own weight of water, submerging the fraction 750/1000 of its volume). The net force is zero, so the apparent weight is 0 N. The value 1.5 N subtracts only three quarters of the weight, but the whole weight is already supported by the upthrust.

62. How much heat must be supplied to 2.0 kg of water to raise its temperature from 20 °C to 70 °C? (c of water = 4200 J/(kg K); ignore heat losses.)

- A)** 84 kJ **B)** 210 kJ **C)** 294 kJ **D)** 420 kJ **E)** 588 kJ

Answer: D

$Q = mc\Delta T$ with $\Delta T = 70 - 20 = 50 \text{ K}$: $Q = 2.0 \times 4200 \times 50 = 420\,000 \text{ J} = 420 \text{ kJ}$

The value 588 kJ uses the final temperature 70 °C instead of the 50 K rise, and 210 kJ corresponds to only 1.0 kg of water.

63. A 0.40 kg block of aluminium at 100 °C is dropped into 0.60 kg of water at 20 °C in an insulated container. (c of aluminium = 900 J/(kg K), c of water = 4200 J/(kg K).) What is the final equilibrium temperature?

- A)** 20 °C **B)** 30 °C **C)** 40 °C **D)** 52 °C **E)** 60 °C

Answer: B

Heat lost by aluminium = heat gained by water: $0.40 \times 900 \times (100 - T) = 0.60 \times 4200 \times (T - 20)$.

So $36\,000 - 360T = 2520T - 50\,400$, hence $86\,400 = 2880T$ and $T = 30\text{ °C}$

The value 52 °C is the mass-weighted average of the starting temperatures, ignoring that water's specific heat is far larger than aluminium's; 60 °C is the plain average.

64. In an insulated cup, 0.50 kg of water at 12 °C is mixed with hot water at 82 °C, and the mixture settles at 32 °C. (c of water = 4200 J/(kg K); ignore heat losses.) What mass of hot water was added?

- A)** 0.10 kg **B)** 0.20 kg **C)** 0.35 kg **D)** 0.50 kg **E)** 0.70 kg

Answer: B

Same substance on both sides, so c cancels: $m_{\text{cold}} \times \Delta T_{\text{cold}} = m_{\text{hot}} \times \Delta T_{\text{hot}}$.

Thus $0.50 \times (32 - 12) = m \times (82 - 32)$, so $10 = 50m$ and $m = 0.20\text{ kg}$

The value 0.70 kg uses the 70 K gap between the two starting temperatures instead of the cold water's actual 20 K rise.

65. An electric heater consumes 2.5 kWh of energy. (1 kWh = 3.6 MJ.) This energy, expressed in megajoules, is:

- A)** 0.90 MJ **B)** 2.5 MJ **C)** 9.0 MJ **D)** 25 MJ **E)** 90 MJ

Answer: C

Each kWh equals 3.6 MJ, so $2.5\text{ kWh} = 2.5 \times 3.6 = 9.0\text{ MJ}$

The value 2.5 MJ treats 1 kWh as 1 MJ and drops the 3.6 factor.

66. A heating element connected across a 12 V supply carries a current of 2.0 A. How much energy does it convert to heat in 5.0 minutes?

- A)** 0.60 kJ **B)** 0.72 kJ **C)** 3.6 kJ **D)** 7.2 kJ **E)** 72 kJ

Answer: D

Power = $VI = 12 \times 2.0 = 24\text{ W}$; time = $5.0 \times 60 = 300\text{ s}$

Energy = $Pt = 24 \times 300 = 7200\text{ J} = 7.2\text{ kJ}$

The value 3.6 kJ multiplies voltage by time and forgets the current, and 0.72 kJ treats 5.0 minutes as 30 s.

67. Two identical $6.0\ \Omega$ resistors are connected, first in series and then in parallel, across a $12\ \text{V}$ battery of negligible internal resistance. Which one of the following statements is false?

- A)** The parallel arrangement dissipates 4 times as much total power as the series arrangement.
- B)** In the parallel arrangement the battery supplies $48\ \text{W}$.
- C)** In the series arrangement each resistor dissipates $6.0\ \text{W}$.
- D)** In the series arrangement the battery supplies $12\ \text{W}$.
- E)** The equivalent resistance of the parallel arrangement is $6.0\ \Omega$.

Answer: E

Series: total resistance $12\ \Omega$, current $1.0\ \text{A}$, so the battery supplies $12\ \text{W}$ and each resistor dissipates $I^2R = 6.0\ \text{W}$, both true. Parallel: equivalent resistance $(6.0 \times 6.0)/(6.0 + 6.0) = 3.0\ \Omega$, so the battery supplies $12^2 \div 3.0 = 48\ \text{W}$, exactly four times the series power, also true. The false statement is the one giving the parallel equivalent resistance as $6.0\ \Omega$: two equal resistors in parallel halve the resistance rather than leaving it unchanged.

68. A $6.0\ \Omega$ resistor is connected in parallel with a $3.0\ \Omega$ resistor, and this pair is connected in series with a $5.0\ \Omega$ resistor across a supply. What is the equivalent resistance of the whole network?

- A)** $2.0\ \Omega$ **B)** $5.0\ \Omega$ **C)** $7.0\ \Omega$ **D)** $11\ \Omega$ **E)** $14\ \Omega$

Answer: C

Parallel pair: $(6.0 \times 3.0)/(6.0 + 3.0) = 2.0\ \Omega$

In series with $5.0\ \Omega$: total = $2.0 + 5.0 = 7.0\ \Omega$

The value $14\ \Omega$ adds all three as if all were in series, and $2.0\ \Omega$ forgets the series resistor.

69. A battery of negligible internal resistance supplies $24\ \text{V}$ to an $8.0\ \Omega$ resistor connected in series with a parallel combination of a $12\ \Omega$ resistor and a $6.0\ \Omega$ resistor. What current does the battery supply?

- A)** $1.5\ \text{A}$ **B)** $2.0\ \text{A}$ **C)** $2.4\ \text{A}$ **D)** $3.0\ \text{A}$ **E)** $6.0\ \text{A}$

Answer: B

Parallel pair: $(12 \times 6.0)/(12 + 6.0) = 4.0\ \Omega$

Total resistance = $8.0 + 4.0 = 12\ \Omega$, so $I = 24 \div 12 = 2.0\ \text{A}$

The value $3.0\ \text{A}$ divides $24\ \text{V}$ by the $8.0\ \Omega$ alone, ignoring the parallel section, and $1.5\ \text{A}$ doubles the total resistance.

70. A $6.0\ \Omega$ resistor draws $2.0\ \text{A}$ from a battery of negligible internal resistance. When a second resistor is connected in parallel with it, the current drawn from the battery rises to $3.0\ \text{A}$. What is the resistance of the second resistor?

- A)** $3.0\ \Omega$ **B)** $6.0\ \Omega$ **C)** $12\ \Omega$ **D)** $18\ \Omega$ **E)** $36\ \Omega$

Answer: C

Battery voltage: $V = 6.0 \times 2.0 = 12\ \text{V}$

Adding a parallel branch does not change the voltage, so the original resistor still draws $2.0\ \text{A}$ and the extra $3.0 - 2.0 = 1.0\ \text{A}$ flows through the new branch.

Its resistance is $12 \div 1.0 = 12\ \Omega$

The value $6.0\ \Omega$ assumes the extra current is shared equally between the branches, and $18\ \Omega$ adds resistances as if the second resistor were in series.

71. A $4.0\ \Omega$ resistor and a $12\ \Omega$ resistor are connected in series across a $6.0\ \text{V}$ battery of negligible internal resistance. A wire of negligible resistance is then connected across the $12\ \Omega$ resistor. What current does the battery supply once the wire is connected?

- A)** $0.375\ \text{A}$ **B)** $0.50\ \text{A}$ **C)** $1.5\ \text{A}$ **D)** $3.0\ \text{A}$ **E)** $6.0\ \text{A}$

Answer: C

The wire of negligible resistance short-circuits the $12\ \Omega$ resistor, so only the $4.0\ \Omega$ resistor limits the current: $I = 6.0 \div 4.0 = 1.5\ \text{A}$. The value $0.375\ \text{A}$ is the current before the wire is added ($6.0\ \text{V}$ across $4.0 + 12 = 16\ \Omega$), and $0.50\ \text{A}$ would follow from short-circuiting the $4.0\ \Omega$ instead of the $12\ \Omega$.

72. Intensity measures how fast a wave delivers energy to a surface, per unit area of that surface. The SI unit in which intensity is expressed is:

- A)** W/m^3 **B)** J/m^2 **C)** W/m **D)** J/s **E)** W/m^2

Answer: E

Intensity = power per unit area = energy $\div$ (time $\times$ area). Since power in watts is joules per second, the unit is W/m^2 . The unit J/s is the watt alone, with no area division, and J/m^2 is energy per area with the per-second factor missing.

73. A small lamp is treated as a point source radiating uniformly in all directions. At $2.0\ \text{m}$ from the lamp the intensity of its radiation is $36\ \text{W/m}^2$. What is the intensity at $6.0\ \text{m}$ from the lamp?

- A)** $4.0\ \text{W/m}^2$ **B)** $6.0\ \text{W/m}^2$ **C)** $12\ \text{W/m}^2$ **D)** $18\ \text{W/m}^2$ **E)** $36\ \text{W/m}^2$

Answer: A

A point source spreads its power over a sphere of area $4\pi r^2$, so intensity follows an inverse-square law. Tripling the distance from $2.0\ \text{m}$ to $6.0\ \text{m}$ divides the intensity by $3^2 = 9$: $36 \div 9 = 4.0\ \text{W/m}^2$. The value $12\ \text{W/m}^2$ scales as $1/r$ instead of $1/r^2$.

74. In a heated room, a radiator warms the air beside it. This air rises, cooler air replaces it, and the resulting circulating flow carries thermal energy around the room. This circulating flow transports the energy mainly by:

- A)** conduction through the metal of the radiator, the best heat conductor present
B) evaporation, because the flow carries water vapour that releases heat where it condenses
C) radiation, because the hot radiator emits infrared rays that travel in straight lines
D) conduction, because neighbouring air molecules pass energy on by direct collisions
E) convection, because the warmed air expands, becomes less dense and rises within the flow

Answer: E

Energy carried by the bulk motion of a warmed fluid is convection: the heated air expands, becomes less dense, rises, and cooler air sinks to replace it, forming the circulating flow that transports the energy. Conduction passes energy between neighbouring molecules without bulk flow, and radiation travels as electromagnetic waves, not as moving air.

75. A vacuum flask keeps a drink hot. It has double walls with silvered surfaces and a vacuum between them, closed at the top by a stopper. Which statement about how the flask limits heat loss is correct?

- A)** The silvered walls mainly reduce conduction through the walls.
- B)** The vacuum blocks convection only, because conduction can still occur through a vacuum.
- C)** The stopper works mainly by reflecting radiation back into the drink.
- D)** The vacuum mainly blocks radiation, and the silvering mainly blocks convection.
- E)** The vacuum blocks both conduction and convection, and the silvered surfaces cut radiation losses.

Answer: E

Conduction and convection both need a material medium, so the vacuum blocks both paths at once. The silvered surfaces reflect infrared radiation back towards the drink, cutting radiation losses. Saying the vacuum blocks only convection is wrong because conduction has no medium in a vacuum either, and it is the silvering, not the vacuum, that handles radiation.

Section G - Logic and argument analysis (Q76-95)

76. In a library, every manuscript in the special archive is catalogued. Some of the archive's manuscripts are illustrated, and no illustrated item in the archive may leave the building. Which of the following must be true?

- A)** Some of the archive's illustrated manuscripts are catalogued.
- B)** Every catalogued item in the library is illustrated.
- C)** Every item that may leave the building is catalogued.
- D)** Some catalogued items in the library may leave the building.
- E)** No catalogued item in the archive is illustrated.

Answer: A

Premise two guarantees that at least one illustrated archive manuscript exists, and premise one catalogues every archive manuscript, so at least one manuscript is both illustrated and catalogued: some of the archive's illustrated manuscripts are catalogued must be true. The claim that every catalogued item is illustrated is the invalid converse of the first premise. Items that may leave the building need not be catalogued, because non-archive items fall outside the cataloguing rule, and denying that any catalogued archive item is illustrated simply contradicts the premises.

77. At a tennis club, only members with a current season pass may book the courts, and all club rules are followed. Nadia booked a court this week. Some members with a season pass also take coaching lessons. Which of the following must be true?

- A)** Nadia holds a current season pass.
- B)** Every member who books the courts takes coaching lessons.
- C)** Some members who book the courts take coaching lessons.
- D)** Every member with a season pass books the courts.
- E)** Nadia takes coaching lessons.

Answer: A

'Only members with a pass may book' means booking implies holding a pass. Nadia booked a court, so she holds a current season pass. The claim that every pass holder books the courts merely reverses the rule, and the coaching-lesson statements concern pass holders who need not be Nadia.

78. Every swimmer who won a medal this season trains at the Riverside pool, and nobody who trains at the Riverside pool pays a joining fee. Which of the following must be true?

- A)** Some people who train at the Riverside pool did not win a medal this season.
- B)** The Riverside pool charges no joining fee to anyone.
- C)** Some swimmers who pay a joining fee won a medal this season.
- D)** Every swimmer who trains at the Riverside pool won a medal this season.
- E)** No swimmer who won a medal this season pays a joining fee.

Answer: E

Chain the two universals: every medalling swimmer trains at Riverside, and nobody who trains at Riverside pays a joining fee, so no swimmer who won a medal pays a joining fee. The claims that every Riverside trainer medalled, or that fee-paying swimmers medalled, reverse or contradict the chain, and the premises say nothing about fees for people who do not train at Riverside.

79. Some players in the squad are left-footed. No left-footed player takes the corner kicks, and every player in the squad is a professional. Which of the following must be true?

- A)** Some professionals do not take the corner kicks.
- B)** Every player who takes the corner kicks is right-footed.
- C)** No professional takes the corner kicks.
- D)** Every player who takes the corner kicks is in the squad.
- E)** Some left-footed players are not professionals.

Answer: A

At least one squad player is left-footed, and every squad player is a professional, so at least one professional is left-footed. No left-footed player takes the corners, so that professional does not take them: some professionals do not take the corner kicks. The claim that every corner taker is right-footed adds an assumption, since the premises never say that not left-footed means right-footed.

80. Every baker at the Saturday market arrives before dawn, and some bakers at the market sell rye bread. Which of the following must be true?

- A)** Everyone who arrives before dawn sells rye bread.
- B)** Some people who arrive before dawn sell rye bread.
- C)** All sellers of rye bread at the market are bakers.
- D)** No baker at the market sells rye bread.
- E)** Some bakers at the market arrive after dawn.

Answer: B

Some bakers sell rye bread, and every baker arrives before dawn, so at least one of the before-dawn arrivals sells rye bread. The claim that everyone arriving before dawn sells rye reverses the first premise, and nothing makes rye sellers bakers in general.

81. Every member of the rowing club can swim, and no member of the rowing club is under sixteen. Tara is a member of the rowing club. Which of the following must be true?

- A)** Some people under sixteen can swim.
- B)** Everyone who can swim is a member of the rowing club.
- C)** Tara is under sixteen.
- D)** Only swimmers may join the rowing club.
- E)** Tara can swim and is not under sixteen.

Answer: E

Both premises are universal and Tara is a member, so both apply to her: she can swim and she is not under sixteen. The claim that every swimmer is a club member reverses the first premise, and the premises say nothing about who is allowed to join.

82. About a postponed football match, Ria says: 'The match was postponed because of rain.' Tom says: 'The match was postponed because of rain, and Ria is lying.' Exactly one of the two always tells the truth and the other always lies. Which of the following must be true?

- A)** Tom tells the truth and rain caused the postponement.
- B)** Ria tells the truth and rain caused the postponement.
- C)** Ria tells the truth and the postponement had another cause.
- D)** Tom tells the truth and the postponement had another cause.
- E)** Both of them always tell the truth.

Answer: B

Test Tom as the truth teller: his statement then asserts that rain caused the postponement and that Ria is lying. But Ria asserted exactly that rain caused it, so she would be telling the truth, a contradiction. Hence Tom is the liar and Ria the truth teller, which also fixes the fact that rain caused the postponement; Tom's conjunction fails because its second half is false.

83. Three pupils, Amara, Ben and Cara, are asked about a missing lab key. Each of them either always tells the truth or always lies. Amara says: 'Ben is lying.' Ben says: 'Cara is lying.' Cara says: 'Both Amara and Ben are lying.' Which of the following is correct?

- A)** Amara and Cara lie and Ben tells the truth.
- B)** The three statements allow more than one consistent assignment.
- C)** Ben and Cara lie and Amara tells the truth.
- D)** Amara and Ben tell the truth and Cara lies.
- E)** All three tell the truth.

Answer: A

Suppose Amara tells the truth: then Ben lies, so Ben's claim is false and Cara tells the truth; yet Cara's claim that both Amara and Ben lie would then be false, which a truth teller cannot utter. So Amara lies, making Ben's claim true and Ben truthful; Cara's claim that both Amara and Ben lie is then false, fitting a liar. The assignment is unique: Amara and Cara lie, Ben tells the truth.

84. Three teachers are questioned about a borrowed stopwatch. Exactly one of them is lying and the other two are telling the truth. Priya says: 'I did not take the stopwatch.' Quinn says: 'Priya is lying.' Ravi says: 'Quinn is lying.' Which of the following must be true?

- A)** Priya took the stopwatch.
- B)** All three are telling the truth.
- C)** Quinn took the stopwatch.
- D)** Ravi took the stopwatch.
- E)** Priya did not take the stopwatch.

Answer: E

Quinn and Ravi contradict each other, so the single liar must be one of that pair; either way Priya is among the two truth tellers and her denial is accurate. If Priya were the liar, Quinn and Ravi would both be truthful, but they contradict, which is impossible. So Priya did not take the stopwatch, although the statements never settle who actually did, which is why the options naming a culprit fail.

85. After a match, the scoreboard was found broken. Exactly one of the three candidates for team captain always tells the truth and the other two always lie. Dana says: 'Eli broke the scoreboard.' Eli says: 'I did not break the scoreboard.' Fran says: 'Dana is lying.' Which of the following must be true?

- A)** Eli tells the truth and Fran broke the scoreboard.
- B)** Fran tells the truth and Dana broke the scoreboard.
- C)** Eli tells the truth and Dana broke the scoreboard.
- D)** Dana tells the truth and Eli broke the scoreboard.
- E)** Fran tells the truth and Eli broke the scoreboard.

Answer: D

If Eli told the truth, Dana and Fran would both be liars, but Fran's claim that Dana is lying would then be true, impossible for a liar. If Fran told the truth, Dana would be lying, which makes Eli's denial true as well, giving two truth tellers. So Dana is the single truth teller, Eli's denial is false, and Eli broke the scoreboard.

86. At a lost property office, exactly one of Pia's two statements is true and exactly one of Sam's two statements is true. The clerk confirms the blue bag was collected on exactly one day. Pia says: 'The bag was collected on Monday.' and 'The bag was not collected on Wednesday.' Sam says: 'The bag was collected on Wednesday.' and 'The bag was not collected on Monday.' Which of the following must be true?

- A)** The bag was collected on Wednesday.
- B)** The bag was collected on Monday.
- C)** Sam made two true statements.
- D)** Pia made two false statements.
- E)** The bag was collected on a day that is neither Monday nor Wednesday.

Answer: E

If the day were Monday, both of Pia's statements would be true; if it were Wednesday, both would be false. Neither fits her requirement of exactly one true statement. Any other day makes 'not Wednesday' true and 'Monday' false for Pia, and likewise leaves Sam with exactly one true statement, so the collection happened on a day that is neither Monday nor Wednesday.

87. A notice at the swimming pool reads: 'If the pool is at capacity, new swimmers are turned away.' Which of the following is logically equivalent to the notice?

- A)** If the pool is not at capacity, new swimmers are turned away.
- B)** If new swimmers are not turned away, the pool is not at capacity.
- C)** If new swimmers are turned away, the pool is at capacity.
- D)** If the pool is not at capacity, new swimmers are not turned away.
- E)** New swimmers are turned away only if the pool is at capacity.

Answer: B

The notice reads: if at capacity, then turned away. A conditional is equivalent only to its contrapositive: if new swimmers are not turned away, the pool is not at capacity. The version reading 'if turned away, then at capacity' merely reverses the conditional, and 'if not at capacity, then not turned away' denies both parts in the same order, which is not equivalent.

88. The coach tells the squad: 'If a player attends every training session this month, she will be selected for the final.' Which of the following is logically equivalent to the coach's statement?

- A)** A player who attends at least one training session this month will be selected for the final.
- B)** A player will be selected for the final only if she attends every training session this month.
- C)** If a player does not attend every training session this month, she will not be selected for the final.
- D)** If a player is not selected for the final, she did not attend every training session this month.
- E)** If a player is selected for the final, she attended every training session this month.

Answer: D

The coach's promise is: perfect attendance $\rightarrow$ selected. Its contrapositive, 'not selected $\rightarrow$ attendance was not perfect', is the only equivalent form. The mirror statement 'imperfect attendance $\rightarrow$ not selected' is the inverse and does not follow, and 'selected $\rightarrow$ perfect attendance' merely reverses the promise.

89. Under the library's loan rule, a book may be borrowed for a full term only if it is on the reserve list. Which of the following must also be true if the rule is obeyed?

- A)** A book not on the reserve list may not be borrowed for a full term.
- B)** Every book on the reserve list is borrowed for a full term.
- C)** A book not on the reserve list may be borrowed for a full term.
- D)** A book on the reserve list may be borrowed for a full term.
- E)** Only books on the reserve list may be borrowed at all.

Answer: A

'Full term only if on the reserve list' means full-term borrowing implies a reserve listing. Denying the requirement gives the equivalent claim: a book not on the reserve list may not be borrowed for a full term. A reserve listing alone does not grant full-term borrowing (that is the converse), and the rule says nothing about banning shorter loans.

90. A ferry company's policy states that if a sailing is delayed, meal vouchers are issued, and that if meal vouchers are issued, the terminal canteen stays open late. Which of the following is logically equivalent to the policy taken as a whole?

- A)** Every delayed sailing leaves passengers hungry.
- B)** If a sailing is delayed, it is because the canteen plans to stay open late.
- C)** If the canteen does not stay open late, the sailing was not delayed.
- D)** If a sailing is not delayed, the canteen does not stay open late.
- E)** If the canteen stays open late, the sailing was delayed.

Answer: C

Chaining the rules: delayed $\rightarrow$ vouchers $\rightarrow$ canteen open late, so 'delayed $\rightarrow$ canteen late'. Its contrapositive, 'canteen does not stay open late $\rightarrow$ the sailing was not delayed', is equivalent to the policy as a whole. Starting from no delay, or inferring a delay from a late canteen, reverses the direction of the chain and is not entailed.

91. A security manual states: 'The side gate stays locked if and only if the alarm is armed.' Which of the following is logically equivalent to this statement?

- A)** The alarm is armed whenever the side gate is unlocked.
- B)** If the alarm is disarmed, the side gate is locked.
- C)** If the alarm is armed, the side gate is unlocked.
- D)** If the side gate is locked, the alarm is disarmed.
- E)** The side gate is unlocked exactly when the alarm is disarmed.

Answer: E

A biconditional ('locked if and only if armed') stays true when both sides are negated together: the gate is unlocked exactly when the alarm is disarmed. The statements pairing unlocked with armed, or locked with disarmed, contradict the manual, and mixing a negated side with the plain other side breaks the two-way match.

92. A council reports that streets repaved with a new porous surface drain rainwater quickly and form far fewer puddles, and concludes that repaving will reduce road accidents in which vehicles slip. The conclusion depends on assuming which of the following?

- A)** Accidents in which vehicles slip are usually caused by surface water, and the porous surface does not make roads slippery in other ways.
- B)** Drivers on repaved streets tend to be more cautious than other drivers.
- C)** The porous surface costs less per kilometre than standard asphalt.
- D)** The council intends to repave most streets within five years.
- E)** Standard asphalt forms more puddles than the porous surface.

Answer: A

The argument runs from fewer puddles to fewer slip accidents, which needs two things: that surface water is what usually makes vehicles slip, and that the porous surface introduces no new slipperiness of its own. The statement combining exactly those two ideas supplies the missing link. Costs and council intentions are beside the chain, driver caution would be a rival explanation rather than a needed assumption, and the asphalt comparison merely restates a premise.

93. At a gym, members on the flexible plan attend twice as often as members on the standard plan and report larger fitness gains. The manager concludes that the flexible plan itself causes the better fitness. Which of the following, if true, most weakens the manager's conclusion?

- A)** The flexible plan costs more per month than the standard plan.
- B)** Members choose their own plan, and most members who were already training hard before joining picked the flexible plan.
- C)** Standard plan members report busier weekday schedules.
- D)** Attendance at the gym is recorded automatically at the entrance.
- E)** Both groups took the same fitness assessment at the start and at the end of the study.

Answer: B

If members choose their own plans and the members already training hard mostly picked the flexible plan, then the higher attendance and gains existed before the plan and need not be caused by it. This self-selection explanation strikes directly at the causal claim. The higher fee and the automatic recording do not touch the cause, and the identical assessments for both groups strengthen the study rather than weaken it.

94. A ferry operator installed automatic ticket scanners on two of its six routes. Boarding time on those two routes fell by a third within two months and stayed flat on the other four. The operator concludes the scanners caused the faster boarding. Which of the following, if true, most strengthens this conclusion?

- A)** Passengers on the scanner routes said in a survey that they liked the scanners.
- B)** Both scanner routes use ferries of the same model.
- C)** Over the same two months the operator ran an advertising campaign on all six routes.
- D)** When the scanners were out of service for two days, boarding times on those routes briefly returned to their old level and fell again after repairs.
- E)** Staff numbers were also increased at the terminals of the two scanner routes.

Answer: D

The strongest support is the withdrawal test: boarding times returned exactly to their old level while the scanners were out of service and fell again after repairs, reconnecting the effect to the suspected cause far more tightly than the before-and-after contrast alone. Extra staff on the same two routes suggest a rival cause, the advertising campaign covered all six routes so it cannot explain the difference between them, and passenger approval says nothing about speed.

95. Last year a sports centre cut its evening swim sessions from five per week to three. Over the year, total attendance at evening swims stayed almost unchanged, and the waiting list for evening swims grew longer. Which of the following is most strongly supported by this information?

- A)** Some swimmers gave up swimming in the evenings.
- B)** The centre should restore the two cancelled sessions.
- C)** On average, each remaining evening session now accommodates more swimmers than before.
- D)** The waiting list caused the sessions to be cut.
- E)** Attendance at morning swims must have increased.

Answer: C

Total evening attendance stayed roughly constant while the sessions fell from five to three per week, so the same total was spread over fewer sessions: on average each remaining session now accommodates more swimmers. Nothing in the data shows that anyone gave up evening swimming, that morning sessions changed, or that the waiting list existed before the cuts, and what the centre should do is a recommendation the data cannot settle.